enРБНФ №1 **(опис синтаксису всіма допустимими засобами РБНФ)**	**РБНФ №2** **(опис формальної граматики засобами РБНФ)**	**Формальна граматика**	**Формальна граматика з специфікацією lookahead у правилах для LL(2)-аналізатора**	/* **Перевірка РБНФ №1 за допомогою коду** *(помістити у файл "EBNF_N1.h")* */	/* **Перевірка РБНФ №2 за допомогою коду** *(помістити у файл "EBNF_N2.h")* */	/* **Перевірка прототипу LL(2)-синтаксичного аналізатора (спеціальна структура) та прототипу лексичного аналізатора (регулярні вирази) за допомогою коду.** *Лексеми для синтаксичного аналізатора обробляються лексичним аналізатором, тому синтаксичний аналізатор не аналізує їх повторно (як показано в РБНФ).* *(помістити у файл "LexicaByRegExAndSyntaxByLL2prototototype.h")* УВАГА: при копіюванні зважайте, щоб у кожному рядку після символу «\» не містилось жодних інших символів. */
		G = (N, T, P, S)	G = (N, T, P, S)			
		S → program_rule	S → program_rule			
		N = { labeled_point, goto_label, program_name, value_type, array_specify, declaration_element, array_specify__optional, other_declaration_ident, declaration, other_declaration_ident__iteration, index_action, unary_operator, unary_operation, binary_operator, binary_action, left_expression, group_expression, index_action__optional, expression, binary_action__iteration, expression_or_cond_block__with_optional_assign, assign_to_right, assign_to_right__optional, if_expression, body_for_true, false_cond_block_without_else, body_for_false, cond_block, false_cond_block_without_else__iteration, body_for_false__optional, cycle_begin_expression, cycle_end_expression, cycle_counter, cycle_counter_lr_init, cycle_counter_init, cycle_counter_last_value, cycle_body, forto_direction, forto_cycle, continue_while, break_while, statement_in_while_and_if_body, statement, block_statements_in_while_and_if_body, statement_in_while_and_if_body__iteration, while_cycle_head_expression, while_cycle, repeat_until_cycle_cond, repeat_until_cycle, statements__or__block_statements, block_statements, input_rule, argument_for_input, output_rule, statement__iteration, expression__optional, program_rule, declaration__optional, non_zero_digit, digit__iteration, digit, unsigned_value, value, sign__optional, sign, ident, letter_in_upper_case, letter_in_lower_case, sign_plus, sign_minus }	N = { labeled_point, goto_label, program_name, value_type, array_specify, declaration_element, array_specify__optional, other_declaration_ident, declaration, other_declaration_ident__iteration, index_action, unary_operator, unary_operation, binary_operator, binary_action, left_expression, group_expression, index_action__optional, expression, binary_action__iteration, expression_or_cond_block__with_optional_assign, assign_to_right, assign_to_right__optional, if_expression, body_for_true, false_cond_block_without_else, body_for_false, cond_block, false_cond_block_without_else__iteration, body_for_false__optional, cycle_begin_expression, cycle_end_expression, cycle_counter, cycle_counter_lr_init, cycle_counter_init, cycle_counter_last_value, cycle_body, forto_direction, forto_cycle, continue_while, break_while, statement_in_while_and_if_body, statement, block_statements_in_while_and_if_body, statement_in_while_and_if_body__iteration, while_cycle_head_expression, while_cycle, repeat_until_cycle_cond, repeat_until_cycle, statements__or__block_statements, block_statements, input_rule, argument_for_input, output_rule, statement__iteration, expression__optional, program_rule, declaration__optional, non_zero_digit, digit__iteration, digit, unsigned_value, value, sign__optional, sign, ident, letter_in_upper_case, letter_in_lower_case, sign_plus, sign_minus }	#define NONTERMINALS labeled_point, \ goto_label, \ program_name, \ value_type, \ array_specify, \ declaration_element, \ \ other_declaration_ident, \ declaration, \ \ index_action, \ unary_operator, \ unary_operation, \ binary_operator, \ binary_action, \ left_expression, \ group_expression, \ \ expression, \ \ expression_or_cond_block__with_optional_assign, \ assign_to_right, \ \ if_expression, \ body_for_true, \ false_cond_block_without_else, \ body_for_false, \ cond_block, \ \ \ cycle_begin_expression, \ cycle_end_expression, \ cycle_counter, \ cycle_counter_lr_init, \ cycle_counter_init, \ cycle_counter_last_value, \ cycle_body, \ forto_direction, \ forto_cycle, \ continue_while, \ break_while, \ statement_in_while_and_if_body, \ statement, \ block_statements_in_while_and_if_body, \ \ while_cycle_head_expression, \ while_cycle, \ repeat_until_cycle_cond, \ repeat_until_cycle, \ statements__or__block_statements, \ block_statements, \ input_rule, \ argument_for_input, \ output_rule, \ \ \ program_rule, \ \ non_zero_digit, \ digit__iteration, \ digit, \ unsigned_value, \ value, \ \ sign, \ ident, \ letter_in_upper_case, \ letter_in_lower_case, \ sign_plus, \ sign_minus	#define NONTERMINALS labeled_point, \ goto_label, \ program_name, \ value_type, \ array_specify, \ declaration_element, \ array_specify__optional, \ other_declaration_ident, \ declaration, \ other_declaration_ident__iteration, \ index_action, \ unary_operator, \ unary_operation, \ binary_operator, \ binary_action, \ left_expression, \ group_expression, \ index_action__optional, \ expression, \ binary_action__iteration, \ expression_or_cond_block__with_optional_assign, \ assign_to_right, \ assign_to_right__optional, \ if_expression, \ body_for_true, \ false_cond_block_without_else, \ body_for_false, \ cond_block, \ false_cond_block_without_else__iteration, \ body_for_false__optional, \ cycle_begin_expression, \ cycle_end_expression, \ cycle_counter, \ cycle_counter_lr_init, \ cycle_counter_init, \ cycle_counter_last_value, \ cycle_body, \ forto_direction, \ forto_cycle, \ continue_while, \ break_while, \ statement_in_while_and_if_body, \ statement, \ block_statements_in_while_and_if_body, \ statement_in_while_and_if_body__iteration, \ while_cycle_head_expression, \ while_cycle, \ repeat_until_cycle_cond, \ repeat_until_cycle, \ statements__or__block_statements, \ block_statements, \ input_rule, \ argument_for_input, \ output_rule, \ statement__iteration, \ expression__optional, \ program_rule, \ declaration__optional, \ non_zero_digit, \ digit__iteration, \ digit, \ unsigned_value, \ value, \ sign__optional, \ sign, \ ident, \ letter_in_upper_case, \ letter_in_lower_case, \ sign_plus, \ sign_minus	
		T = { ":" , "GOTO" ,	T = { ":" , "GOTO" ,	#define TOKENS \ tokenCOLON, \ tokenGOTO, \	#define TOKENS \ tokenCOLON, \ tokenGOTO, \	

<table>
<tr><td></td><td></td><td>"INTEGER16" ,
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<tr><td></td><td></td><td></td><td></td><td>#define COMMENT_BEGIN_STR "#*"
#define COMMENT_END_STR "*#"</td><td>#define COMMENT_BEGIN_STR "#*"
#define COMMENT_END_STR "*#"</td><td>#define COMMENT_BEGIN_STR "#*"
#define COMMENT_END_STR "*#"</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td><td>#define TOKENS_RE ";|:=|=:|\\+|-|*|,|==|!=|:|\\[|\\]|\\(|\\)|\\{|\\}|<=|>=|[_0-9A-Za-z]+|[^ \t\r\f\v\n]"</td></tr>
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<table>
<tr><td></td><td></td><td></td><td></td><td></td><td></td><td>#define KEYWORDS_RE ";|:=|=:|\\+|-
|*|,|==|!=|:|\\[|\\]|\\(|\\)|\\{|\\}|NAME|DATA|BODY|END|BREAK|CONTINUE|GET|PUT|IF|ELSE|FOR|TO|DOWNTO|DO|WHILE|REPEAT|UNTIL|GOTO|DIV|MOD|<=|>=|NOT|AND|OR|INTEGER16"</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td><td>#define IDENTIFIERS_RE "_[A-Z][A-Z][A-Z][A-Z][A-Z][A-Z][A-Z]"</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td><td>#define UNSIGNEDVALUES_RE "0|[1-9][0-9]*"</td></tr>
<tr><td></td><td></td><td></td><td></td><td>tokenGROUPEXPRESSIONBEGIN = "(" >> BOUNDARIES;</td><td>tokenGROUPEXPRESSIONBEGIN = "(" >> BOUNDARIES;</td><td>#define T_BEGIN_GROUPEXPRESSION_0 "("
#define T_BEGIN_GROUPEXPRESSION_1 ""
#define T_BEGIN_GROUPEXPRESSION_2 ""
#define T_BEGIN_GROUPEXPRESSION_3 ""</td></tr>
<tr><td></td><td></td><td></td><td></td><td>tokenGROUPEXPRESSIONEND = ")" >> BOUNDARIES;</td><td>tokenGROUPEXPRESSIONEND = ")" >> BOUNDARIES;</td><td>#define T_END_GROUPEXPRESSION_0 ")"
#define T_END_GROUPEXPRESSION_1 ""
#define T_END_GROUPEXPRESSION_2 ""
#define T_END_GROUPEXPRESSION_3 ""</td></tr>
<tr><td></td><td></td><td></td><td></td><td>tokenLEFTSQUAREBRACKETS = "[" >> BOUNDARIES;</td><td>tokenLEFTSQUAREBRACKETS = "[" >> BOUNDARIES;</td><td>#define T_LEFT_SQUAREBRACKETS_0 "["
#define T_LEFT_SQUAREBRACKETS_1 ""
#define T_LEFT_SQUAREBRACKETS_2 ""
#define T_LEFT_SQUAREBRACKETS_3 ""</td></tr>
<tr><td></td><td></td><td></td><td></td><td>tokenRIGHTSQUAREBRACKETS = "]" >> BOUNDARIES;</td><td>tokenRIGHTSQUAREBRACKETS = "]" >> BOUNDARIES;</td><td>#define T_RIGHT_SQUAREBRACKETS_0 "]"
#define T_RIGHT_SQUAREBRACKETS_1 ""
#define T_RIGHT_SQUAREBRACKETS_2 ""
#define T_RIGHT_SQUAREBRACKETS_3 ""</td></tr>
<tr><td></td><td></td><td></td><td></td><td>tokenBEGINBLOCK = "{" >> BOUNDARIES;</td><td>tokenBEGINBLOCK = "{" >> BOUNDARIES;</td><td>#define T_BEGIN_BLOCK_0 "{"
#define T_BEGIN_BLOCK_1 ""
#define T_BEGIN_BLOCK_2 ""
#define T_BEGIN_BLOCK_3 ""</td></tr>
<tr><td></td><td></td><td></td><td></td><td>tokenENDBLOCK = "}" >> BOUNDARIES;</td><td>tokenENDBLOCK = "}" >> BOUNDARIES;</td><td>#define T_END_BLOCK_0 "}"
#define T_END_BLOCK_1 ""
#define T_END_BLOCK_2 ""
#define T_END_BLOCK_3 ""</td></tr>
<tr><td></td><td></td><td></td><td></td><td>tokenSEMICOLON = ";" >> BOUNDARIES;</td><td>tokenSEMICOLON = ";" >> BOUNDARIES;</td><td>#define T_SEMICOLON_0 ";"
#define T_SEMICOLON_1 ""
#define T_SEMICOLON_2 ""
#define T_SEMICOLON_3 ""</td></tr>
<tr><td></td><td></td><td></td><td></td><td>tokenCOLON = ":" >> BOUNDARIES;</td><td>tokenCOLON = ":" >> BOUNDARIES;</td><td>#define T_COLON_0 ":"
#define T_COLON_1 ""
#define T_COLON_2 ""
#define T_COLON_3 ""</td></tr>
<tr><td></td><td></td><td></td><td></td><td>tokenGOTO = "GOTO" >> STRICT_BOUNDARIES;</td><td>tokenGOTO = "GOTO" >> STRICT_BOUNDARIES;</td><td>#define T_GOTO_0 "GOTO"
#define T_GOTO_1 ""
#define T_GOTO_2 ""
#define T_GOTO_3 ""</td></tr>
<tr><td></td><td></td><td></td><td></td><td>tokenINTEGER16 = "INTEGER16" >> STRICT_BOUNDARIES;</td><td>tokenINTEGER16 = "INTEGER16" >>
STRICT_BOUNDARIES;</td><td>#define T_DATA_TYPE_0 "INTEGER16"
#define T_DATA_TYPE_1 ""
#define T_DATA_TYPE_2 ""
#define T_DATA_TYPE_3 ""</td></tr>
<tr><td></td><td></td><td></td><td></td><td>tokenCOMMA = "," >> BOUNDARIES;</td><td>tokenCOMMA = "," >> BOUNDARIES;</td><td>#define T_COMA_0 ","
#define T_COMA_1 ""
#define T_COMA_2 ""
#define T_COMA_3 ""</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td><td>#define T_BITWISE_NOT_0 "~"
#define T_BITWISE_NOT_1 ""
#define T_BITWISE_NOT_2 ""
#define T_BITWISE_NOT_3 ""</td></tr>
<tr><td></td><td></td><td></td><td></td><td>tokenNOT = "NOT" >> STRICT_BOUNDARIES;</td><td>tokenNOT = "NOT" >> STRICT_BOUNDARIES;</td><td>#define T_NOT_0 "NOT"
#define T_NOT_1 ""
#define T_NOT_2 ""
#define T_NOT_3 ""</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td><td>#define T_BITWISE_AND_0 "&"
#define T_BITWISE_AND_1 ""
#define T_BITWISE_AND_2 ""
#define T_BITWISE_AND_3 ""</td></tr>
<tr><td></td><td></td><td></td><td></td><td>tokenAND = "AND" >> STRICT_BOUNDARIES;</td><td>tokenAND = "AND" >> STRICT_BOUNDARIES;</td><td>#define T_AND_0 "AND"
#define T_AND_1 ""
#define T_AND_2 ""
#define T_AND_3 ""</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td><td>#define T_BITWISE_OR_0 "|"
#define T_BITWISE_OR_1 ""
#define T_BITWISE_OR_2 ""
#define T_BITWISE_OR_3 ""</td></tr>
<tr><td></td><td></td><td></td><td></td><td>tokenOR = "OR" >> STRICT_BOUNDARIES;</td><td>tokenOR = "OR" >> STRICT_BOUNDARIES;</td><td>#define T_OR_0 "OR"
#define T_OR_1 ""
#define T_OR_2 ""
#define T_OR_3 ""</td></tr>
<tr><td></td><td></td><td></td><td></td><td>tokenEQUAL = "==" >> BOUNDARIES;</td><td>tokenEQUAL = "==" >> BOUNDARIES;</td><td>#define T_EQUAL_0 "=="
#define T_EQUAL_1 ""
#define T_EQUAL_2 ""
#define T_EQUAL_3 ""</td></tr>
<tr><td></td><td></td><td></td><td></td><td>tokenNOTEQUAL = "!=" >> BOUNDARIES;</td><td>tokenNOTEQUAL = "!=" >> BOUNDARIES;</td><td>#define T_NOT_EQUAL_0 "!="
#define T_NOT_EQUAL_1 ""
#define T_NOT_EQUAL_2 ""
#define T_NOT_EQUAL_3 ""</td></tr>
<tr><td></td><td></td><td></td><td></td><td>tokenLESSOREQUAL = "<=" >> BOUNDARIES;</td><td>tokenLESSOREQUAL = "<=" >> BOUNDARIES;</td><td>#define T_LESS_OR_EQUAL_0 "<="
#define T_LESS_OR_EQUAL_1 ""
#define T_LESS_OR_EQUAL_2 ""
#define T_LESS_OR_EQUAL_3 ""</td></tr>
<tr><td></td><td></td><td></td><td></td><td>tokenGREATEROREQUAL = ">=" >> BOUNDARIES;</td><td>tokenGREATEROREQUAL = ">=" >> BOUNDARIES;</td><td>#define T_GREATER_OR_EQUAL_0 ">="
#define T_GREATER_OR_EQUAL_1 ""
#define T_GREATER_OR_EQUAL_2 ""
#define T_GREATER_OR_EQUAL_3 ""</td></tr>
<tr><td></td><td></td><td></td><td></td><td>tokenPLUS = "+" >> BOUNDARIES;</td><td>tokenPLUS = "+" >> BOUNDARIES;</td><td>#define T_ADD_0 "+"
#define T_ADD_1 ""
#define T_ADD_2 ""
#define T_ADD_3 ""</td></tr>
<tr><td></td><td></td><td></td><td></td><td>tokenMINUS = "-" >> BOUNDARIES;</td><td>tokenMINUS = "-" >> BOUNDARIES;</td><td>#define T_SUB_0 "-"
#define T_SUB_1 ""
#define T_SUB_2 ""
#define T_SUB_3 ""</td></tr>
<tr><td></td><td></td><td></td><td></td><td>tokenMUL = "*" >> BOUNDARIES;</td><td>tokenMUL = "*" >> BOUNDARIES;</td><td>#define T_MUL_0 "*"
#define T_MUL_1 ""
#define T_MUL_2 ""
#define T_MUL_3 ""</td></tr>
<tr><td></td><td></td><td></td><td></td><td>tokenDIV = "DIV" >> STRICT_BOUNDARIES;</td><td>tokenDIV = "DIV" >> STRICT_BOUNDARIES;</td><td>#define T_DIV_0 "DIV"
#define T_DIV_1 ""
#define T_DIV_2 ""
#define T_DIV_3 ""</td></tr>
<tr><td></td><td></td><td></td><td></td><td>tokenMOD = "MOD" >> STRICT_BOUNDARIES;</td><td>tokenMOD = "MOD" >> STRICT_BOUNDARIES;</td><td>#define T_MOD_0 "MOD"
#define T_MOD_1 ""</td></tr>
</table>

						#define T_MOD_2 "" #define T_MOD_3 ""
				tokenLRASSIGN = "=:" >> BOUNDARIES;	tokenLRASSIGN = "=:" >> BOUNDARIES;	#define T_LRASSIGN_0 "=:" #define T_LRASSIGN_1 "" #define T_LRASSIGN_2 "" #define T_LRASSIGN_3 ""
						#define T_THEN_BLOCK_0 "{" #define T_THEN_BLOCK_1 "" #define T_THEN_BLOCK_2 "" #define T_THEN_BLOCK_3 ""
				tokenELSE = "ELSE" >> STRICT_BOUNDARIES;	tokenELSE = "ELSE" >> STRICT_BOUNDARIES;	#define T_ELSE_BLOCK_0 "ELSE" #define T_ELSE_BLOCK_1 T_BEGIN_BLOCK_0 #define T_ELSE_BLOCK_2 "" #define T_ELSE_BLOCK_3 ""
				tokenIF = "IF" >> STRICT_BOUNDARIES;	tokenIF = "IF" >> STRICT_BOUNDARIES;	#define T_IF_0 "IF" #define T_IF_1 "" #define T_IF_2 "" #define T_IF_3 ""
						#define T_ELSE_IF_0 "ELSE" #define T_ELSE_IF_1 T_IF_0 #define T_ELSE_IF_2 "" #define T_ELSE_IF_3 ""
				tokenDO = "DO" >> STRICT_BOUNDARIES;	tokenDO = "DO" >> STRICT_BOUNDARIES;	#define T_DO_0 "DO" #define T_DO_1 "" #define T_DO_2 "" #define T_DO_3 ""
				tokenFOR = "FOR" >> STRICT_BOUNDARIES;	tokenFOR = "FOR" >> STRICT_BOUNDARIES;	#define T_FOR_0 "FOR" #define T_FOR_1 "" #define T_FOR_2 "" #define T_FOR_3 ""
				tokenTO = "TO" >> STRICT_BOUNDARIES;	tokenTO = "TO" >> STRICT_BOUNDARIES;	#define T_TO_0 "TO" #define T_TO_1 "" #define T_TO_2 "" #define T_TO_3 ""
				tokenDOWNTO = "DOWNTO" >> STRICT_BOUNDARIES;	tokenDOWNTO = "DOWNTO" >> STRICT_BOUNDARIES;	#define T_DOWNTO_0 "DOWNTO" #define T_DOWNTO_1 "" #define T_DOWNTO_2 "" #define T_DOWNTO_3 ""
				tokenWHILE = "WHILE" >> STRICT_BOUNDARIES;	tokenWHILE = "WHILE" >> STRICT_BOUNDARIES;	#define T_WHILE_0 "WHILE" #define T_WHILE_1 "" #define T_WHILE_2 "" #define T_WHILE_3 ""
				tokenCONTINUE = "CONTINUE" >> STRICT_BOUNDARIES;	tokenCONTINUE = "CONTINUE" >> STRICT_BOUNDARIES;	#define T_CONTINUE_WHILE_0 "CONTINUE" #define T_CONTINUE_WHILE_1 "" #define T_CONTINUE_WHILE_2 "" #define T_CONTINUE_WHILE_3 ""
				tokenBREAK = "BREAK" >> STRICT_BOUNDARIES;	tokenBREAK = "BREAK" >> STRICT_BOUNDARIES;	#define T_EXIT_WHILE_0 "BREAK" #define T_EXIT_WHILE_1 "" #define T_EXIT_WHILE_2 "" #define T_EXIT_WHILE_3 ""
				tokenEXIT = "EXIT" >> STRICT_BOUNDARIES;	tokenEXIT = "EXIT" >> STRICT_BOUNDARIES;	#define T_EXIT_0 "EXIT" #define T_EXIT_1 "" #define T_EXIT_2 "" #define T_EXIT_3 ""
				tokenREPEAT = "REPEAT" >> STRICT_BOUNDARIES;	tokenREPEAT = "REPEAT" >> STRICT_BOUNDARIES;	#define T_REPEAT_0 "REPEAT" #define T_REPEAT_1 "" #define T_REPEAT_2 "" #define T_REPEAT_3 ""
				tokenUNTIL = "UNTIL" >> STRICT_BOUNDARIES;	tokenUNTIL = "UNTIL" >> STRICT_BOUNDARIES;	#define T_UNTIL_0 "UNTIL" #define T_UNTIL_1 "" #define T_UNTIL_2 "" #define T_UNTIL_3 ""
				tokenGET = "GET" >> STRICT_BOUNDARIES;	tokenGET = "GET" >> STRICT_BOUNDARIES;	#define T_INPUT_0 "GET" #define T_INPUT_1 "" #define T_INPUT_2 "" #define T_INPUT_3 ""
				tokenPUT = "PUT" >> STRICT_BOUNDARIES;	tokenPUT = "PUT" >> STRICT_BOUNDARIES;	#define T_OUTPUT_0 "PUT" #define T_OUTPUT_1 "" #define T_OUTPUT_2 "" #define T_OUTPUT_3 ""
				tokenNAME = "NAME" >> STRICT_BOUNDARIES;	tokenNAME = "NAME" >> STRICT_BOUNDARIES;	#define T_NAME_0 "NAME" #define T_NAME_1 "" #define T_NAME_2 "" #define T_NAME_3 ""
				tokenBODY = "BODY" >> STRICT_BOUNDARIES;	tokenBODY = "BODY" >> STRICT_BOUNDARIES;	#define T_BODY_0 "BODY" #define T_BODY_1 "" #define T_BODY_2 "" #define T_BODY_3 ""
				tokenDATA = "DATA" >> STRICT_BOUNDARIES;	tokenDATA = "DATA" >> STRICT_BOUNDARIES;	#define T_DATA_0 "DATA" #define T_DATA_1 "" #define T_DATA_2 "" #define T_DATA_3 ""
				tokenBEGIN = "BEGIN" >> STRICT_BOUNDARIES;	tokenBEGIN = "BEGIN" >> STRICT_BOUNDARIES;	#define T_BEGIN_0 "BEGIN" #define T_BEGIN_1 "" #define T_BEGIN_2 "" #define T_BEGIN_3 ""
				tokenEND = "END" >> STRICT_BOUNDARIES;	tokenEND = "END" >> STRICT_BOUNDARIES;	#define T_END_0 "END" #define T_END_1 "" #define T_END_2 "" #define T_END_3 ""
						#define T_NULL_STATEMENT_0 "NULL" #define T_NULL_STATEMENT_1 "STATEMENT" #define T_NULL_STATEMENT_2 "" #define T_NULL_STATEMENT_3 ""
						#define GRAMMAR_LL2__2025 {\
labeled_point = ident , ":";	labeled_point = ident , ":";	labeled_point → ident ":"	labeled_point(1: "ident_terminal") → ident ":"	labeled_point = ident >> tokenCOLON;	labeled_point = ident >> tokenCOLON;	{ LA_IS, {"ident_terminal"}, { "labeled_point",{\ { LA_IS, {""}, 2, {"ident", T_COLON_0}}\ }}},\
goto_label = "GOTO" , ident;	goto_label = "GOTO" , ident;	goto_label → "GOTO" ident	goto_label(1: "GOTO") → "GOTO" ident	goto_label = tokenGOTO >> ident;	goto_label = tokenGOTO >> ident;	{ LA_IS, {T_GOTO_0}, { "goto_label",{\ { LA_IS, {""}, 2, {T_GOTO_0, "ident"}}\ }}},\
program_name = ident;	program_name = ident;	program_name → ident	program_name(1: "ident_terminal") → ident	program_name = SAME_RULE(ident);	program_name = SAME_RULE(ident);	{ LA_IS, {"ident_terminal"}, { "program_name",{\ { LA_IS, {""}, 1, {"ident"}}\ }}},\
value_type = "INTEGER16";	value_type = "INTEGER16";	value_type → "INTEGER16"	value_type(1: "INTEGER16") → "INTEGER16"	value_type = SAME_RULE(tokenINTEGER16);	value_type = SAME_RULE(tokenINTEGER16);	{ LA_IS, {T_DATA_TYPE_0}, { "value_type",{\ { LA_IS, {""}, 1, {T_DATA_TYPE_0}}\

<table>
<tr><td></td><td></td><td></td><td></td><td></td><td></td><td>}}},\</td></tr>
<tr><td></td><td>array_specify = "[", unsigned_value, "]";</td><td>array_specify → "[" unsigned_value "]"</td><td>array_specify(1: "[") → "[" unsigned_value "]"</td><td></td><td>array_specify = "[" >> unsigned_value >> "]";</td><td>{ LA_IS, {"["}, { "array_specify",{\
{ LA_IS, {""}, 3, {"[", "unsigned_value", "]"}}\
}}},\</td></tr>
<tr><td>declaration_element = ident, ["[", unsigned_value , "]"];</td><td>declaration_element = ident, array_specify__optional;</td><td>declaration_element → ident array_specify__optional</td><td>declaration_element(1: "ident_terminal") → ident array_specify__optional</td><td>declaration_element = ident >> -(tokenLEFTSQUAREBRACKETS >> unsigned_value >> tokenRIGHTSQUAREBRACKETS);</td><td>declaration_element = ident >> array_specify__optional;</td><td>{ LA_IS, {"ident_terminal"}, { "declaration_element",{\
{ LA_IS, {""}, 2, {"ident", "array_specify__optional"}}\
}}},\</td></tr>
<tr><td></td><td>array_specify__optional = array_specify | ε;</td><td>array_specify__optional → array_specify
array_specify__optional → ε</td><td>array_specify__optional(1: "[") → array_specify
array_specify__optional(1: !"[") → ε</td><td></td><td>array_specify__optional = array_specify | "";</td><td>{ LA_IS, {"["}, { "array_specify__optional",{\
{ LA_IS, {""}, 1, {"array_specify"}}\
}}},\
{ LA_NOT, {"["}, { "array_specify__optional",{\
{ LA_IS, {""}, 0, {""}}\
}}},\</td></tr>
<tr><td>other_declaration_ident = "," , declaration_element;</td><td>other_declaration_ident = "," , declaration_element;</td><td>other_declaration_ident → "," declaration_element</td><td>other_declaration_ident(1: ",") → "," declaration_element</td><td>other_declaration_ident = tokenCOMMA >> declaration_element;</td><td>other_declaration_ident = tokenCOMMA >> declaration_element;</td><td>{ LA_IS, {T_COMA_0}, { "other_declaration_ident",{\
{ LA_IS, {""}, 2, {T_COMA_0, "declaration_element"}}\
}}},\</td></tr>
<tr><td>declaration = value_type , declaration_element , {other_declaration_ident};</td><td>declaration = value_type , declaration_element , other_declaration_ident__iteration;</td><td>declaration → value_type declaration_element other_declaration_ident__iteration</td><td>declaration(1: "INTEGER16") → value_type declaration_element other_declaration_ident__iteration</td><td>declaration = value_type >> declaration_element >> *other_declaration_ident;</td><td>declaration = value_type >> declaration_element >> other_declaration_ident__iteration;</td><td>{ LA_IS, {T_DATA_TYPE_0}, { "declaration",{\
{ LA_IS, {""}, 3, {"value_type", "declaration_element", "other_declaration_ident__iteration"}}\
}}},\</td></tr>
<tr><td></td><td>other_declaration_ident__iteration = other_declaration_ident, other_declaration_ident__iteration | ε;</td><td>other_declaration_ident__iteration → other_declaration_ident other_declaration_ident__iteration
false_cond_block_without_else__iteration → ε</td><td>other_declaration_ident__iteration(1: ",") → other_declaration_ident other_declaration_ident__iteration
false_cond_block_without_else__iteration(1: !",") → ε</td><td></td><td>other_declaration_ident__iteration = other_declaration_ident >> other_declaration_ident__iteration | "";</td><td>{ LA_IS, { T_COMA_0 }, { "other_declaration_ident__iteration",{\
{ LA_IS, {""}, 2, { "other_declaration_ident", "other_declaration_ident__iteration" }}\
}}},\
{ LA_NOT, { T_COMA_0 }, { "other_declaration_ident__iteration",{\
{ LA_IS, {""}, 0, { "" }}\
}}},\</td></tr>
<tr><td>index_action = "[" , expression , "]";</td><td>index_action = "[" , expression , "]";</td><td>index_action → "[" expression "]"</td><td>index_action(1: "[") → "[" expression "]"</td><td>index_action = tokenLEFTSQUAREBRACKETS >> expression >> tokenRIGHTSQUAREBRACKETS;</td><td>index_action = tokenLEFTSQUAREBRACKETS >> expression >> tokenRIGHTSQUAREBRACKETS;</td><td>{ LA_IS, { "[" }, { "index_action",{\
{ LA_IS, {""}, 3, { "[", "expression", "]" }}\
}}},\</td></tr>
<tr><td>unary_operator = "NOT";</td><td>unary_operator = "NOT";</td><td>unary_operator → "NOT"</td><td>unary_operator(1: "NOT") → "NOT"</td><td>unary_operator = SAME_RULE(tokenNOT);</td><td>unary_operator = SAME_RULE(tokenNOT);</td><td>{ LA_IS, { T_NOT_0 }, { "unary_operator",{\
{ LA_IS, {""}, 1, { T_NOT_0 }}\
}}},\</td></tr>
<tr><td>unary_operation = unary_operator , expression;</td><td>unary_operation = unary_operator , expression;</td><td>unary_operation → unary_operator expression</td><td>unary_operation(1: "NOT") → unary_operator expression</td><td>unary_operation = unary_operator >> expression;</td><td>unary_operation = unary_operator >> expression;</td><td>{ LA_IS, { T_NOT_0 }, { "unary_operation",{\
{ LA_IS, {""}, 2, { "unary_operator", "expression" }}\
}}},\</td></tr>
<tr><td>binary_operator = "AND" | "OR" | "==" | "!=" | "<=" | ">=" | "+" | "-" | "*" | "DIV" | "MOD";</td><td>binary_operator = "AND" | "OR" | "==" | "!=" | "<=" | ">=" | "+" | "-" | "*" | "DIV" | "MOD";</td><td>binary_operator → "AND"
binary_operator → "OR"
binary_operator → "=="
binary_operator → "!="
binary_operator → "<="
binary_operator → ">="
binary_operator → "+"
binary_operator → "-"
binary_operator → "*"
binary_operator → "DIV"
binary_operator → "MOD"</td><td>binary_operator(1: "AND") → "AND"
binary_operator(1: "OR") → "OR"
binary_operator(1: "==") → "=="
binary_operator(1: "!=") → "!="
binary_operator(1: "<=") → "<="
binary_operator(1: ">=") → ">="
binary_operator(1: "+") → "+"
binary_operator(1: "-") → "-"
binary_operator(1: "*") → "*"
binary_operator(1: "DIV") → "DIV"
binary_operator(1: "MOD") → "MOD"</td><td>binary_operator = tokenAND | tokenOR | tokenEQUAL | tokenNOTEQUAL | tokenLESSOREQUAL | tokenGREATEROREQUAL | tokenPLUS | tokenMINUS | tokenMUL | tokenDIV | tokenMOD;</td><td>binary_operator = tokenAND | tokenOR | tokenEQUAL | tokenNOTEQUAL | tokenLESSOREQUAL | tokenGREATEROREQUAL | tokenPLUS | tokenMINUS | tokenMUL | tokenDIV | tokenMOD;</td><td>{ LA_IS, { T_AND_0 }, { "binary_operator",{\
{ LA_IS, {""}, 1, { T_AND_0 }}\
}}},\
{ LA_IS, { T_OR_0 }, { "binary_operator",{\
{ LA_IS, {""}, 1, { T_OR_0 }}\
}}},\
{ LA_IS, { T_EQUAL_0 }, { "binary_operator",{\
{ LA_IS, {""}, 1, { T_EQUAL_0 }}\
}}},\
{ LA_IS, { T_NOT_EQUAL_0 }, { "binary_operator",{\
{ LA_IS, {""}, 1, { T_NOT_EQUAL_0 }}\
}}},\
{ LA_IS, { T_LESS_OR_EQUAL_0 }, { "binary_operator",{\
{ LA_IS, {""}, 1, { T_LESS_OR_EQUAL_0 }}\
}}},\
{ LA_IS, { T_GREATER_OR_EQUAL_0 }, { "binary_operator",{\
{ LA_IS, {""}, 1, { T_GREATER_OR_EQUAL_0 }}\
}}},\
{ LA_IS, { T_ADD_0 }, { "binary_operator",{\
{ LA_IS, {""}, 1, { T_ADD_0 }}\
}}},\
{ LA_IS, { T_SUB_0 }, { "binary_operator",{\
{ LA_IS, {""}, 1, { T_SUB_0 }}\
}}},\
{ LA_IS, { T_MUL_0 }, { "binary_operator",{\
{ LA_IS, {""}, 1, { T_MUL_0 }}\
}}},\
{ LA_IS, { T_DIV_0 }, { "binary_operator",{\
{ LA_IS, {""}, 1, { T_DIV_0 }}\
}}},\
{ LA_IS, { T_MOD_0 }, { "binary_operator",{\
{ LA_IS, {""}, 1, { T_MOD_0 }}\
}}},\</td></tr>
<tr><td>binary_action = binary_operator , expression;</td><td>binary_action = binary_operator , expression;</td><td>binary_action → binary_operator expression</td><td>binary_action(1: "AND", "OR", "==", "!=", "<=", ">=", "+", "-", "*", "DIV", "MOD") → binary_operator expression</td><td>binary_action = binary_operator >> expression;</td><td>binary_action = binary_operator >> expression;</td><td>{ LA_IS, { T_AND_0, T_OR_0, T_EQUAL_0, T_NOT_EQUAL_0, T_LESS_OR_EQUAL_0, T_GREATER_OR_EQUAL_0, T_ADD_0, T_SUB_0, T_MUL_0, T_DIV_0, T_MOD_0 }, { "binary_action",{\
{ LA_IS, {""}, 2, { "binary_operator", "expression" }}\
}}},\</td></tr>
<tr><td>left_expression = group_expression | unary_operation | ident , [index_action] | value | cond_block;</td><td>left_expression = group_expression | unary_operation | ident , index_action__optional | value | cond_block;</td><td>left_expression → group_expression
left_expression → unary_operation
left_expression → ident , index_action__optional
left_expression → value
left_expression → cond_block</td><td>left_expression(1: "(") → group_expression
left_expression(1: "NOT") → unary_operation
left_expression(1: "_") → ident , index_action__optional
left_expression(1: "0", "1", "2", "3", "4", "5", "6", "7", "8", "9") → value
left_expression(1: "+", "-"; 2: "0", "1", "2", "3", "4", "5", "6", "7", "8", "9") → value
left_expression(1: "IF") → cond_block</td><td>left_expression = group_expression | unary_operation | ident >> -index_action | value | cond_block;</td><td>left_expression = group_expression | unary_operation | ident >> index_action__optional | value | cond_block;</td><td>{LA_IS, { "(" }, { "left_expression",{\
{LA_IS, { "" }, 1, { "group_expression" }}\
}}},\
{LA_IS, { T_NOT_0 }, { "left_expression",{\
{LA_IS, { "" }, 1, { "unary_operation" }}\
}}},\
{LA_IS, { "ident_terminal" }, { "left_expression",{\
{LA_IS, {""}, 2, { "ident", "index_action__optional" }}\
}}},\
{LA_IS, { "unsigned_value_terminal" }, { "left_expression",{\
{LA_IS, {""}, 1, { "value" }}\
}}},\
{LA_IS, { T_ADD_0, T_SUB_0 }, { "left_expression",{\
{LA_IS, { "unsigned_value_terminal"}, 1, { "value" }},\
/*{LA_NOT, { "unsigned_value_terminal" }, 1, { "unary_operation" }}*/\
}}},\
{LA_IS, { T_IF_0 }, { "left_expression",{\
{LA_IS, { "" }, 1, { "cond_block" }}\
}}},\</td></tr>
<tr><td></td><td>index_action__optional = index_action | ε;</td><td>index_action__optional → index_action
index_action__optional → ε</td><td>index_action__optional(1: "[") → index_action
index_action__optional(1: !"[") → ε</td><td></td><td>index_action__optional = index_action | "";</td><td>{LA_IS, { "[" }, { "index_action__optional",{\
{LA_IS, {""}, 1, { "index_action" }}\
}}},\
{LA_NOT, { "[" }, { "index_action__optional",{\
{LA_IS, {""}, 0, { "" }}\
}}},\</td></tr>
<tr><td>expression = left_expression , {binary_action};</td><td>expression = left_expression , binary_action__iteration;</td><td>expression → left_expression binary_action__iteration</td><td>expression(1: "(", "NOT", "+", "-", "_", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "IF") →</td><td>expression = left_expression >> *binary_action;</td><td>expression = left_expression >> binary_action__iteration;</td><td>{LA_IS, { "(", T_NOT_0, T_ADD_0, T_SUB_0, "ident_terminal", "unsigned_value_terminal", T_IF_0 },</td></tr>
</table>

			left_expression binary_action__iteration			{ "expression",{\ {LA_IS, {""}, 2, { "left_expression", "binary_action__iteration" }}\ }}},\
	binary_action__iteration = binary_action, binary_action__iteration | ε;	binary_action__iteration → binary_action binary_action__iteration binary_action__iteration → ε	binary_action__iteration(1: "AND", "OR", "==", "!=", "<=", ">=", "+", "-", "*", "DIV", "MOD") → binary_action binary_action__iteration binary_action__iteration(1: !"AND", !"OR", !"==", !"!=", !"<=", !">=", !"+", !"-", !"*", !"DIV", !"MOD") → ε		binary_action__iteration = binary_action >> binary_action__iteration | "";	{LA_IS, { T_AND_0, T_OR_0, T_EQUAL_0, T_NOT_EQUAL_0, T_LESS_OR_EQUAL_0, T_GREATER_OR_EQUAL_0, T_ADD_0, T_SUB_0, T_MUL_0, T_DIV_0, T_MOD_0 }, { "binary_action__iteration",{\ {LA_IS, {""}, 2, { "binary_action", "binary_action__iteration" }}\ }}},\ {LA_NOT, { T_AND_0, T_OR_0, T_EQUAL_0, T_NOT_EQUAL_0, T_LESS_OR_EQUAL_0, T_GREATER_OR_EQUAL_0, T_ADD_0, T_SUB_0, T_MUL_0, T_DIV_0, T_MOD_0 }, { "binary_action__iteration",{\ {LA_IS, {""}, 0, { "" }}\ }}},\
group_expression = "(" , expression , ")";	group_expression = "(" , expression , ")";	group_expression → "(" expression ")"	group_expression(1: "(") → "(" expression ")"	group_expression = tokenGROUPEXPRESSIONBEGIN >> expression >> tokenGROUPEXPRESSIONEND;	group_expression = tokenGROUPEXPRESSIONBEGIN >> expression >> tokenGROUPEXPRESSIONEND;	{LA_IS, { "(" }, { "group_expression",{\ {LA_IS, {""}, 3, { "(", "expression", ")" }}\ }}},\
expression_or_cond_block__with_optional_assign = expression , [":=" , ident , [index_action]];	expression_or_cond_block__with_optional_assign = expression , assign_to_right__optional;	expression_or_cond_block__with_optional_assign → expression assign_to_right__optional	expression_or_cond_block__with_optional_assign(1: "(", "NOT", "+", "-", "_", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "IF") → expression assign_to_right__optional	expression_or_cond_block__with_optional_assign = expression >> -(tokenLRASSIGN >> ident >> -index_action);	expression_or_cond_block__with_optional_assign = expression >> assign_to_right__optional;	{LA_IS, { "(", T_NOT_0, T_ADD_0, T_SUB_0, "ident_terminal", "unsigned_value_terminal", T_IF_0 }, { "expression_or_cond_block__with_optional_assign",{\ {LA_IS, {""}, 2, { "expression", "assign_to_right__optional" }}\ }}},\
	assign_to_right = ":=" , ident , index_action__optional;	assign_to_right → ":=" ident index_action__optional	assign_to_right(1: ":=") → ":=" ident index_action__optional		assign_to_right = tokenLRASSIGN >> ident >> index_action__optional;	{LA_IS, { T_LRASSIGN_0 }, { "assign_to_right",{\ {LA_IS, {""}, 3, { T_LRASSIGN_0, "ident", "index_action__optional" }}\ }}},\

	assign_to_right__optional = assign_to_right | ε;	assign_to_right__optional → assign_to_right
		assign_to_right__optional → ε;
			assign_to_right__optional(1: ":=") → assign_to_right
			assign_to_right__optional(1: !":=") → ε;
					assign_to_right__optional = assign_to_right | "";	{ LA_IS, { T_LRASSIGN_0 }, {
						"assign_to_right__optional",{\
						{ LA_IS, {""}, 1, { "assign_to_right" }}\
						}}},\
						{ LA_NOT, { T_LRASSIGN_0 }, {
						"assign_to_right__optional",{\
						{ LA_IS, {""}, 0, { "" }}\
						}}},\

if_expression = expression;	if_expression = expression;	if_expression → expression	if_expression(1: "(", "NOT", "+", "-", "_", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "IF") → expression	if_expression = SAME_RULE(expression);	if_expression = SAME_RULE(expression);	{LA_IS, { "(", T_NOT_0, T_ADD_0, T_SUB_0, "ident_terminal", "unsigned_value_terminal", T_IF_0 }, { "if_expression",{\ {LA_IS, {""}, 1, { "expression" }}\ }}},\
body_for_true = block_statements_in_while_and_if_body;	body_for_true = block_statements_in_while_and_if_body;	body_for_true → block_statements_in_while_and_if_body	body_for_true(1: "{") → block_statements_in_while_and_if_body	body_for_true = SAME_RULE(block_statements_in_while_and_if_body);	body_for_true = SAME_RULE(block_statements_in_while_and_if_body);	{LA_IS, { T_BEGIN_BLOCK_0 }, { "body_for_true",{\ {LA_IS, {""}, 1, { "block_statements_in_while_and_if_body" }}\ }}},\
false_cond_block_without_else = "ELSE" , "IF" , if_expression , body_for_true;	false_cond_block_without_else = "ELSE" , "IF" , if_expression , body_for_true;	false_cond_block_without_else → "ELSE" "IF" if_expression body_for_true	false_cond_block_without_else(1: "ELSE") → "ELSE" "IF" if_expression body_for_true	false_cond_block_without_else = tokenELSE >> tokenIF >> if_expression >> body_for_true;	false_cond_block_without_else = tokenELSE >> tokenIF >> if_expression >> body_for_true;	{LA_IS, { T_ELSE_IF_0 }, { "false_cond_block_without_else",{\ {LA_IS, {""}, 4, { T_ELSE_IF_0, T_ELSE_IF_1, "if_expression", "body_for_true" }}\ }}},\
body_for_false = "ELSE" , block_statements_in_while_and_if_body;	body_for_false = "ELSE" , block_statements_in_while_and_if_body;	body_for_false → "ELSE" block_statements_in_while_and_if_body	body_for_false(1: "ELSE") → "ELSE" block_statements_in_while_and_if_body	body_for_false = tokenELSE >> block_statements_in_while_and_if_body;	body_for_false = tokenELSE >> block_statements_in_while_and_if_body;	{LA_IS, { T_ELSE_BLOCK_0 }, { "body_for_false",{\ {LA_IS, {""}, 2, { T_ELSE_BLOCK_0, "block_statements" }}\ }}},\
cond_block = "IF" , if_expression , body_for_true, { false_cond_block_without_else } , [body_for_false];	cond_block = "IF" , if_expression , body_for_true, false_cond_block_without_else__iteration , body_for_false__optional;	cond_block → "IF" if_expression body_for_true false_cond_block_without_else__iteration body_for_false__optional	cond_block(1: "IF") → "IF" if_expression body_for_true false_cond_block_without_else__iteration body_for_false__optional	cond_block = tokenIF >> if_expression >> body_for_true >> *false_cond_block_without_else >> -body_for_false;	cond_block = tokenIF >> if_expression >> body_for_true >> false_cond_block_without_else__iteration >> body_for_false__optional;	{LA_IS, { T_IF_0 }, { "cond_block",{\ {LA_IS, {""}, 5, { T_IF_0, "if_expression", "body_for_true", "false_cond_block_without_else__iteration", "body_for_false__optional" }}\ }}},\

	false_cond_block_without_else__iteration =
	false_cond_block_without_else,
	false_cond_block_without_else__iteration | ε;
		false_cond_block_without_else__iteration →
		false_cond_block_without_else
		false_cond_block_without_else__iteration
		false_cond_block_without_else__iteration → ε
			false_cond_block_without_else__iteration(1: "ELSE";
			2: "IF") → false_cond_block_without_else
			false_cond_block_without_else__iteration
			false_cond_block_without_else__iteration(1: "ELSE";
			2: !"IF") → ε
			false_cond_block_without_else__iteration(1:
			!"ELSE") → ε
					false_cond_block_without_else__iteration =
					false_cond_block_without_else >>
					false_cond_block_without_else__iteration | "";
						{LA_IS, { T_ELSE_IF_0 }, {
						"false_cond_block_without_else__iteration",{\
						{LA_IS, {T_ELSE_IF_1}, 2, {
						"false_cond_block_without_else",
						"false_cond_block_without_else__iteration" }},\
						{LA_NOT, { T_ELSE_IF_1 }, 0, { "" }}\
						}}},\
						{LA_NOT, { T_ELSE_IF_0 }, {
						"false_cond_block_without_else__iteration",{\
						{LA_IS, {""}, 0, { "" }}\
						}}},\

	body_for_false__optional = body_for_false | ε;	body_for_false__optional → body_for_false
		body_for_false__optional → ε
			body_for_false__optional(1: "FALSE) →
			body_for_false
			body_for_false__optional(1: !"FALSE") → ε
					body_for_false__optional = body_for_false | "";	{LA_IS, { T_ELSE_BLOCK_0 }, {
						"body_for_false__optional",{\
						{LA_IS, {""}, 1, { "body_for_false" }}\
						}}},\
						{LA_NOT, { T_ELSE_BLOCK_0 }, {
						"body_for_false__optional",{\
						{LA_IS, {""}, 0, { "" }}\
						}}},\

cycle_begin_expression = expression;	cycle_begin_expression = expression;	cycle_begin_expression → expression	cycle_begin_expression(1: "(", "NOT", "+", "-", "_", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "IF") → expression	cycle_begin_expression = SAME_RULE(expression);	cycle_begin_expression = SAME_RULE(expression);	{LA_IS, { "(", T_NOT_0, T_ADD_0, T_SUB_0, "ident_terminal", "unsigned_value_terminal", T_IF_0 }, { "cycle_begin_expression",{\ {LA_IS, {""}, 1, { "expression" }}\ }}},\
cycle_end_expression = expression;	cycle_end_expression = expression;	cycle_end_expression → expression	cycle_end_expression(1: "(", "NOT", "+", "-", "_", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "IF") → expression	cycle_end_expression = SAME_RULE(expression);	cycle_end_expression = SAME_RULE(expression);	{LA_IS, { "(", T_NOT_0, T_ADD_0, T_SUB_0, "ident_terminal", "unsigned_value_terminal", T_IF_0 }, { "cycle_end_expression",{\ {LA_IS, {""}, 1, { "expression" }}\ }}},\
cycle_counter = ident;	cycle_counter = ident;	cycle_counter → ident	cycle_counter(1: "_") → ident	cycle_counter = SAME_RULE(ident);	cycle_counter = SAME_RULE(ident);	{LA_IS, { "ident_terminal" }, { "cycle_counter",{\ {LA_IS, {""}, 1, { "ident" }}\ }}},\
cycle_counter_lr_init = cycle_begin_expression , "=:" , cycle_counter;	cycle_counter_lr_init = cycle_begin_expression , "=:" , cycle_counter;	cycle_counter_lr_init → cycle_begin_expression "=:" cycle_counter	cycle_counter_lr_init(1: "(", "NOT", "+", "-", "_", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "IF") → cycle_begin_expression "=:" cycle_counter	cycle_counter_lr_init = cycle_begin_expression >> tokenLRASSIGN >> cycle_counter;	cycle_counter_lr_init = cycle_begin_expression >> tokenLRASSIGN >> cycle_counter;	{LA_IS, { "(", T_NOT_0, T_ADD_0, T_SUB_0, "ident_terminal", "unsigned_value_terminal", T_IF_0 }, { "cycle_counter_lr_init",{\ {LA_IS, {""}, 3, { "cycle_begin_expression", T_LRASSIGN_0, "cycle_counter" }}\ }}},\
cycle_counter_init = cycle_counter_lr_init;	cycle_counter_init = cycle_counter_lr_init;	cycle_counter_init → cycle_counter_lr_init	cycle_counter_init(1: "(", "NOT", "+", "-", "_", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "IF") → cycle_counter_lr_init	cycle_counter_init = SAME_RULE(cycle_counter_lr_init);	cycle_counter_init = SAME_RULE(cycle_counter_lr_init);	{LA_IS, { "(", T_NOT_0, T_ADD_0, T_SUB_0, "ident_terminal", "unsigned_value_terminal", T_IF_0 }, { "cycle_counter_init",{\ {LA_IS, { "" }, 1, { "cycle_counter_lr_init" }}\ }}},\

<table>
<tr><td>cycle_counter_last_value = cycle_end_expression;</td><td>cycle_counter_last_value = cycle_end_expression;</td><td>cycle_counter_last_value → cycle_end_expression</td><td>cycle_counter_last_value(1: "(", "NOT", "+", "-", "_", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "IF") →
cycle_end_expression</td><td>cycle_counter_last_value = SAME_RULE(cycle_end_expression);</td><td>cycle_counter_last_value =
SAME_RULE(cycle_end_expression);</td><td>{LA_IS, { "(", T_NOT_0, T_ADD_0, T_SUB_0,
"ident_terminal", "unsigned_value_terminal", T_IF_0 },
{ "cycle_counter_last_value",{\
{LA_IS, {""}, 1, { "cycle_end_expression" }}\
}}},\</td></tr>
<tr><td>cycle_body = "DO" , ({statement} | block_statements);</td><td>cycle_body = "DO" ,
statements__or__block_statements;</td><td>cycle_body → "DO"
statements__or__block_statements</td><td>cycle_body(1: "DO") → "DO"
statements__or__block_statements</td><td>cycle_body = tokenDO >> (statement | block_statements);</td><td>cycle_body = tokenDO >>
statements__or__block_statements;</td><td>{LA_IS, { T_DO_0 }, { "cycle_body",{\
{LA_IS, {""}, 2, { T_DO_0,
"statements__or__block_statements" }}\
}}},\</td></tr>
<tr><td></td><td>forto_direction = "TO" | "DOWNTO";</td><td>forto_direction → "TO"
forto_direction → "DOWNTO"</td><td>forto_direction(1: "TO") → "TO"
forto_direction(1: "DOWNTO") → "DOWNTO"</td><td></td><td>forto_direction = tokenTO | tokenDOWNTO;</td><td>{LA_IS, { T_TO_0 }, { "forto_direction",{\
{LA_IS, {""}, 1, { T_TO_0 }}\
}}},\
{LA_IS, { T_DOWNTO_0 }, { "forto_direction",{\
{LA_IS, {""}, 1, { T_DOWNTO_0 }}\
}}},\</td></tr>
<tr><td>forto_cycle = "FOR" , cycle_counter_init , ("TO" | "DOWNTO") , cycle_counter_last_value , cycle_body;</td><td>forto_cycle = "FOR" , cycle_counter_init ,
forto_direction, cycle_counter_last_value ,
cycle_body;</td><td>forto_cycle → "FOR" cycle_counter_init
forto_direction, cycle_counter_last_value
cycle_body</td><td>forto_cycle(1: "FOR") → "FOR" cycle_counter_init
forto_direction, cycle_counter_last_value
cycle_body</td><td>forto_cycle = tokenFOR >> cycle_counter_init >> (tokenTO |
tokenDOWNTO) >> cycle_counter_last_value >> cycle_body;</td><td>forto_cycle = tokenFOR >> cycle_counter_init >>
forto_direction >> cycle_counter_last_value >>
cycle_body;</td><td>{LA_IS, { T_FOR_0 }, { "forto_cycle",{\
{LA_IS, {""}, 5, { T_FOR_0, "cycle_counter_init",
"forto_direction", "cycle_counter_last_value",
"cycle_body" }}\
}}},\</td></tr>
<tr><td></td><td>continue_while = "CONTINUE";</td><td>continue_while → "CONTINUE"</td><td>continue_while(1: "CONTINUE") → "CONTINUE"</td><td>continue_while = SAME_RULE(tokenCONTINUE);</td><td>continue_while = SAME_RULE(tokenCONTINUE);</td><td>{LA_IS, { T_CONTINUE_WHILE_0 }, { "continue_while",{\
{LA_IS, {""}, 1, { T_CONTINUE_WHILE_0 }}\
}}},\</td></tr>
<tr><td></td><td>break_while = "BREAK";</td><td>break_while → "BREAK"</td><td>break_while(1: "BREAK") → "BREAK"</td><td>break_while = SAME_RULE(tokenBREAK);</td><td>break_while = SAME_RULE(tokenBREAK);</td><td>{LA_IS, { T_EXIT_WHILE_0 }, { "break_while",{\
{LA_IS, {""}, 1, { T_EXIT_WHILE_0 }}\
}}},\</td></tr>
<tr><td>statement_in_while_and_if_body = statement | "CONTINUE" | "BREAK";</td><td>statement_in_while_and_if_body = statement |
continue_while | break_while;</td><td>statement_in_while_and_if_body → statement
statement_in_while_and_if_body →
continue_while
statement_in_while_and_if_body → break_while</td><td>statement_in_while_and_if_body(1: "_", "(", "NOT",
"0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "+", "-",
"IF", "FOR", "WHILE", "REPEAT", "GOTO", "GET",
"PUT", ";") → statement
statement_in_while_and_if_body(1: "CONTINUE") →
continue_while
statement_in_while_and_if_body(1: "BREAK") →
break_while</td><td>statement_in_while_and_if_body = statement | continue_while
| break_while;</td><td>statement_in_while_and_if_body = statement |
continue_while | break_while;</td><td>{LA_IS, { "ident_terminal", "(", T_NOT_0,
"unsigned_value_terminal", T_ADD_0, T_SUB_0, T_IF_0,
T_FOR_0, T_WHILE_0, T_REPEAT_0, T_GOTO_0,
T_INPUT_0, T_OUTPUT_0, T_SEMICOLON_0 }, {
"statement_in_while_and_if_body",{\
{LA_IS, {""}, 1, { "statement" }}\
}}},\
{LA_IS, { T_CONTINUE_WHILE_0 }, {
"statement_in_while_and_if_body",{\
{LA_IS, {""}, 1, { "continue_while" }}\
}}},\
{LA_IS, { T_EXIT_WHILE_0 }, {
"statement_in_while_and_if_body",{\
{LA_IS, {""}, 1, { "break_while" }}\
}}},\</td></tr>
<tr><td>block_statements_in_while_and_if_body = "{" , {statement_in_while_and_if_body} , "}";</td><td>block_statements_in_while_and_if_body = "{" ,
statement_in_while_and_if_body__iteration , "}";</td><td>block_statements_in_while_and_if_body → "{"
statement_in_while_and_if_body__iteration "}"</td><td>block_statements_in_while_and_if_body(1: "{") →
"{" statement_in_while_and_if_body__iteration "}"</td><td>block_statements_in_while_and_if_body = tokenBEGINBLOCK
>> *statement_in_while_and_if_body >> tokenENDBLOCK;</td><td>block_statements_in_while_and_if_body =
tokenBEGINBLOCK >>
statement_in_while_and_if_body__iteration >>
tokenENDBLOCK;</td><td>{LA_IS, { T_BEGIN_BLOCK_0 }, {
"block_statements_in_while_and_if_body",{\
{LA_IS, {""}, 3, { T_BEGIN_BLOCK_0,
"statement_in_while_and_if_body__iteration",
T_END_BLOCK_0 }}\
}}},\</td></tr>
<tr><td></td><td>statement_in_while_and_if_body__iteration =
statement_in_while_and_if_body ,
statement_in_while_and_if_body__iteration | ε;</td><td>statement_in_while_and_if_body__iteration →
statement_in_while_and_if_body
statement_in_while_and_if_body__iteration
statement_in_while_and_if_body__iteration → ε</td><td>statement_in_while_and_if_body__iteration(1: "_",
"(", "NOT", "0", "1", "2", "3", "4", "5", "6", "7", "8",
"9", "+", "-", "IF", "FOR", "WHILE", "REPEAT",
"GOTO", "GET", "PUT", ";", "CONTINUE", ";") →
statement_in_while_and_if_body
statement_in_while_and_if_body__iteration
statement_in_while_and_if_body__iteration(1: !"_",
!"(", !"NOT", !"0", !"1", !"2", !"3", !"4", !"5", !"6",
!"7", !"8", !"9",!"+", !"-", !"IF", !"FOR", !"WHILE",
!"REPEAT", !"GOTO", !"GET", !"PUT", !";",
!"CONTINUE", !"BREAK") → ε</td><td></td><td>statement_in_while_and_if_body__iteration =
statement_in_while_and_if_body >>
statement_in_while_and_if_body__iteration | "";</td><td>{LA_IS, { "ident_terminal", "(", T_NOT_0,
"unsigned_value_terminal", T_ADD_0, T_SUB_0, T_IF_0,
T_FOR_0, T_WHILE_0, T_REPEAT_0, T_GOTO_0,
T_INPUT_0, T_OUTPUT_0, T_SEMICOLON_0,
T_CONTINUE_WHILE_0, T_EXIT_WHILE_0 }, {
"statement_in_while_and_if_body__iteration",{\
{LA_IS, {""}, 2, { "statement_in_while_and_if_body",
"statement_in_while_and_if_body__iteration" }}\
}}},\
{LA_NOT, { "ident_terminal", "(", T_NOT_0,
"unsigned_value_terminal", T_ADD_0, T_SUB_0, T_IF_0,
T_FOR_0, T_WHILE_0, T_REPEAT_0, T_GOTO_0,
T_INPUT_0, T_OUTPUT_0, T_SEMICOLON_0,
T_CONTINUE_WHILE_0, T_EXIT_WHILE_0 }, {
"statement_in_while_and_if_body__iteration",{\
{LA_IS, {""}, 0, { "" }}\
}}},\</td></tr>
<tr><td>while_cycle_head_expression = expression;</td><td>while_cycle_head_expression = expression;</td><td>while_cycle_head_expression → expression</td><td>while_cycle_head_expression(1: "(", "NOT", "+", "-",
"_", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "IF")
→ expression</td><td>while_cycle_head_expression = SAME_RULE(expression);</td><td>while_cycle_head_expression =
SAME_RULE(expression);</td><td>{LA_IS, { "(", T_NOT_0, T_ADD_0, T_SUB_0,
"ident_terminal", "unsigned_value_terminal", T_IF_0 },
{ "while_cycle_head_expression",{\
{LA_IS, {""}, 1, { "expression" }}\
}}},\</td></tr>
<tr><td>while_cycle = "WHILE" , while_cycle_head_expression , block_statements_in_while_and_if_body;</td><td>while_cycle = "WHILE" ,
while_cycle_head_expression ,
block_statements_in_while_and_if_body;</td><td>while_cycle → "WHILE"
while_cycle_head_expression
block_statements_in_while_and_if_body</td><td>while_cycle(1: "WHILE") → "WHILE"
while_cycle_head_expression
block_statements_in_while_and_if_body</td><td>while_cycle = tokenWHILE >> while_cycle_head_expression >>
block_statements_in_while_and_if_body;</td><td>while_cycle = tokenWHILE >>
while_cycle_head_expression >>
block_statements_in_while_and_if_body;</td><td>{LA_IS, { T_WHILE_0 }, { "while_cycle",{\
{LA_IS, {""}, 3, { T_WHILE_0,
"while_cycle_head_expression",
"block_statements_in_while_and_if_body" }}\
}}},\</td></tr>
<tr><td>repeat_until_cycle_cond = expression;</td><td>repeat_until_cycle_cond = expression;</td><td>repeat_until_cycle_cond → expression</td><td>repeat_until_cycle_cond(1: "(", "NOT", "+", "-", "_",
"0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "IF") →
expression</td><td>repeat_until_cycle_cond = SAME_RULE(expression);</td><td>repeat_until_cycle_cond = SAME_RULE(expression);</td><td>{LA_IS, { "(", T_NOT_0, T_ADD_0, T_SUB_0,
"ident_terminal", "unsigned_value_terminal", T_IF_0 },
{ "repeat_until_cycle_cond",{\
{LA_IS, {""}, 1, { "expression" }}\
}}},\</td></tr>
<tr><td>repeat_until_cycle = "REPEAT" , ({statement} | block_statements) , "UNTIL" , repeat_until_cycle_cond;</td><td>repeat_until_cycle = "REPEAT" ,
statements__or__block_statements, "UNTIL" ,
repeat_until_cycle_cond;</td><td>repeat_until_cycle → "REPEAT"
statements__or__block_statements "UNTIL"
repeat_until_cycle_cond</td><td>repeat_until_cycle(1: "REPEAT") → "REPEAT"
statements__or__block_statements "UNTIL"
repeat_until_cycle_cond</td><td>repeat_until_cycle = tokenREPEAT >> (*statement |
block_statements) >> tokenUNTIL >> repeat_until_cycle_cond;</td><td>repeat_until_cycle = tokenREPEAT >>
statements__or__block_statements >> tokenUNTIL >>
repeat_until_cycle_cond;</td><td>{LA_IS, { T_REPEAT_0 }, { "repeat_until_cycle",{\
{LA_IS, {""}, 4, { T_REPEAT_0,
"statements__or__block_statements", T_UNTIL_0,
"repeat_until_cycle_cond" }}\
}}},\</td></tr>
<tr><td></td><td>statements__or__block_statements =
statement__iteration | block_statements;</td><td>statements__or__block_statements →
statement__iteration
statements__or__block_statements →
block_statements</td><td>statements__or__block_statements(1: "_", "(",
"NOT", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9",
"+", "-", "IF", "FOR", "WHILE", "REPEAT", "GOTO",
"GET", "PUT", ";") → statement__iteration
statements__or__block_statements(1: "{") →
block_statements</td><td></td><td>statements__or__block_statements =
statement__iteration | block_statements;</td><td>{LA_IS, { "ident_terminal", "(", T_NOT_0,
"unsigned_value_terminal", T_ADD_0, T_SUB_0, T_IF_0,
T_FOR_0, T_WHILE_0, T_REPEAT_0, T_GOTO_0,
T_INPUT_0, T_OUTPUT_0, T_SEMICOLON_0 }, {
"statements__or__block_statements",{\
{LA_IS, {""}, 1, { "statement__iteration" }}\
}}},\
{LA_IS, { T_BEGIN_BLOCK_0 }, {
"statements__or__block_statements",{\
{LA_IS, {""}, 1, { "block_statements" }}\
}}},\</td></tr>
<tr><td>input_rule = "GET" , (ident , [index_action] | "(" , ident , [index_action] , ")");</td><td>input_rule = "GET", argument_for_input;</td><td>input_rule → "GET" argument_for_input</td><td>input_rule(1: "GET") → "GET" argument_for_input</td><td>input_rule = tokenGET >> (ident >> -index_action |
tokenGROUPEXPRESSIONBEGIN >> ident >> -index_action >>
tokenGROUPEXPRESSIONEND);</td><td>input_rule = tokenGET >> argument_for_input;</td><td>{LA_IS, { T_INPUT_0 }, { "input_rule",{\
{LA_IS, {""}, 2, { T_INPUT_0, "argument_for_input" }}\
}}},\</td></tr>
<tr><td></td><td>argument_for_input = ident ,
index_action__optional;
argument_for_input = "(", "ident",
"index_action__optional", ")";</td><td>argument_for_input → ident
index_action__optional
argument_for_input → "(" "ident"
"index_action__optional" ")"</td><td>argument_for_input(1: "_") → ident
index_action__optional
argument_for_input(1: "(") → "(" "ident"
"index_action__optional" ")"</td><td></td><td>argument_for_input = ident >> index_action__optional
| tokenGROUPEXPRESSIONBEGIN >> ident >>
index_action__optional >>
tokenGROUPEXPRESSIONEND;</td><td>{LA_IS, { "ident_terminal" }, { "argument_for_input",{\
{LA_IS, {""}, 2, { "ident", "index_action__optional" }}\
}}},\
{LA_IS, { "(" }, { "argument_for_input",{\
{LA_IS, {""}, 4, { "(", "ident",
"index_action__optional", ")" }}\
}}},\</td></tr>
<tr><td>output_rule = "PUT" , expression;</td><td>output_rule = "PUT", expression;</td><td>output_rule → "PUT" expression</td><td>output(1: "PUT") → "PUT" expression</td><td>output_rule = tokenPUT >> expression;</td><td>output_rule = tokenPUT >> expression;</td><td>{LA_IS, { T_OUTPUT_0 }, { "output_rule", {\
{LA_IS, { "" }, 2, {T_OUTPUT_0, "expression"} }\</td></tr>
</table>

<table>
<tr><td></td><td></td><td></td><td></td><td></td><td></td><td>}}},\</td></tr>
<tr><td>statement = labeled_point | expression_or_cond_block__with_optional_assign | goto_label | forto_cycle | while_cycle | repeat_until_cycle | input_rule | output_rule | ";";</td><td>statement = labeled_point | expression_or_cond_block__with_optional_assign | goto_label | forto_cycle | while_cycle | repeat_until_cycle | input_rule | output_rule | ";";</td><td>statement → labeled_point
statement → expression_or_cond_block__with_optional_assign
statement → goto_label
statement → forto_cycle
statement → while_cycle
statement → repeat_until_cycle
statement → input_rule
statement → output_rule
statement → ";"</td><td>statement(1: "_", ":") → labeled_point
statement(1: "_"; 2: !":") → expression_or_cond_block__with_optional_assign
statement(1: "(", "NOT", "+", "-", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "IF") → expression_or_cond_block__with_optional_assign
statement(1: "GOTO") → goto_label
statement(1: "FOR") → forto_cycle
statement(1: "WHILE") → while_cycle
statement(1: "REPEAT") → repeat_until_cycle
statement(1: "GET") → input_rule
statement(1: "PUT") → output_rule
statement(1: ";") → ";"</td><td>statement = labeled_point /* labeled_point first*/ | expression_or_cond_block__with_optional_assign | goto_label | forto_cycle | while_cycle | repeat_until_cycle | input_rule | output_rule | tokenSEMICOLON;</td><td>statement = labeled_point /* labeled_point first*/ | expression_or_cond_block__with_optional_assign | goto_label | forto_cycle | while_cycle | repeat_until_cycle | input_rule | output_rule | tokenSEMICOLON;</td><td>{LA_IS, { "ident_terminal" }, { "statement", {\
{ LA_IS, { T_COLON_0 }, 1, {"labeled_point"}},\
{ LA_NOT, { T_COLON_0 }, 1, {"expression_or_cond_block__with_optional_assign"}}\
}}},\
{LA_IS, { "(", T_NOT_0, "unsigned_value_terminal", T_ADD_0, T_SUB_0, T_IF_0 }, { "statement", {\
{ LA_IS, {""}, 1, {"expression_or_cond_block__with_optional_assign"}}\
}}},\
{LA_IS, { T_GOTO_0 }, { "statement",{\
{LA_IS, {""}, 1, {"goto_label"}}\
}}},\
{LA_IS, { T_FOR_0 }, { "statement",{\
{LA_IS, {""}, 1, {"forto_cycle"}}\
}}},\
{LA_IS, { T_WHILE_0 }, { "statement",{\
{LA_IS, {""}, 1, {"while_cycle"}}\
}}},\
{LA_IS, { T_REPEAT_0 }, { "statement",{\
{LA_IS, {""}, 1, {"repeat_until_cycle"}}\
}}},\
{LA_IS, { T_INPUT_0 }, { "statement",{\
{LA_IS, {""}, 1, {"input_rule"}}\
}}},\
{LA_IS, { T_OUTPUT_0 }, { "statement",{\
{LA_IS, {""}, 1, {"output_rule"}}\
}}},\
{LA_IS, { T_SEMICOLON_0 }, { "statement",{\
{LA_IS, {""}, 1, {";"}}\
}}},\</td></tr>
<tr><td></td><td>statement__iteration = statement, statement__iteration | ε;</td><td>statement__iteration → statement statement__iteration
statement__iteration → ε</td><td>statement__iteration(1: "_", "(", "NOT", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "+", "-", "IF", "FOR", "WHILE", "REPEAT", "GOTO", "GET", "PUT", ";") → statement statement__iteration
statement__iteration(1: !"_", !"(", !"NOT", !"0", !"1", !"2", !"3", !"4", !"5", !"6", !"7", !"8", !"9", !"+", !"-", !"IF", !"FOR", !"WHILE", !"REPEAT", !"GOTO", !"GET", !"PUT", !";") → ε</td><td></td><td>statement__iteration = statement >> statement__iteration | "";</td><td>{ LA_IS, { "ident_terminal", "(", T_NOT_0, "unsigned_value_terminal", T_ADD_0, T_SUB_0, T_IF_0, T_FOR_0, T_WHILE_0, T_REPEAT_0, T_GOTO_0, T_INPUT_0, T_OUTPUT_0, T_SEMICOLON_0 }, { "statement__iteration",{\
{ LA_IS, {""}, 2, { "statement", "statement__iteration" }}\
}}},\
{ LA_NOT, { "ident_terminal", "(", T_NOT_0, "unsigned_value_terminal", T_ADD_0, T_SUB_0, T_IF_0, T_FOR_0, T_WHILE_0, T_REPEAT_0, T_GOTO_0, T_INPUT_0, T_OUTPUT_0, T_SEMICOLON_0 }, { "statement__iteration",{\
{ LA_IS, {""}, 0, { "" }}\
}}},\</td></tr>
<tr><td>block_statements = "{" , {statement} , "}";</td><td>block_statements = "{" , statement__iteration , "}";</td><td>block_statements → "{" statement__iteration "}"</td><td>block_statements(1: "{") → "{" statement__iteration "}"</td><td>block_statements = tokenBEGINBLOCK >> *statement >> tokenENDBLOCK;</td><td>block_statements = tokenBEGINBLOCK >> statement__iteration >> tokenENDBLOCK;</td><td>{ LA_IS, { T_BEGIN_BLOCK_0 }, { "block_statements",{\
{ LA_IS, {""}, 3, { T_BEGIN_BLOCK_0, "statement__iteration", T_END_BLOCK_0 }}\
}}},\</td></tr>
<tr><td></td><td>expression__optional = expression | "";</td><td>expression__optional → expression
expression__optional → ε</td><td>expression__optional(1: "(", "NOT", "+", "-", "_", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "IF") → expression
expression__optional(1: !"(", !"NOT", !"+", !"-", !"_", !"0", !"1", !"2", !"3", !"4", !"5", !"6", !"7", !"8", !"9", !"IF") → ε</td><td></td><td>expression__optional = expression | "";</td><td>{LA_IS, { "(", T_NOT_0, T_ADD_0, T_SUB_0, "ident_terminal", "unsigned_value_terminal", T_IF_0 }, { "expression__optional",{\
{LA_IS, {""}, 1, { "expression" }}\
}}},\
{LA_NOT, { "(", T_NOT_0, T_ADD_0, T_SUB_0, "ident_terminal", "unsigned_value_terminal", T_IF_0 }, { "expression__optional",{\
{LA_IS, {""}, 0, { "" }}\
}}},\</td></tr>
<tr><td>program_rule = "NAME" , program_name , ";" , "BODY" , "DATA", [declaration] , ";" , {statement} , "END";</td><td>program_rule = "NAME" , program_name , ";" , "BODY" , "DATA", declaration__optional , ";" , statement__iteration , "END";</td><td>program_rule → "NAME" program_name ";" "BODY" "DATA" declaration__optional ";" statement__iteration "END"</td><td>program_rule(1: "NAME") → "NAME" program_name ";" "BODY" "DATA" declaration__optional ";" statement__iteration "END"</td><td>program_rule = BOUNDARIES >> tokenNAME >> program_name >> tokenSEMICOLON >> tokenBODY >> tokenDATA >> (-declaration) >> tokenSEMICOLON >> *statement >> tokenEND;</td><td>program_rule = BOUNDARIES >> tokenNAME >> program_name >> tokenSEMICOLON >> tokenBODY >> tokenDATA >> declaration__optional >> tokenSEMICOLON >> statement__iteration >> tokenEND;</td><td>{ LA_IS, { T_NAME_0 }, { "program_rule",{\
{ LA_IS, {""}, 9, { T_NAME_0, "program_name", T_SEMICOLON_0, T_BODY_0, T_DATA_0, "declaration__optional", T_SEMICOLON_0, "statement__iteration", T_END_0 }}\
}}},\</td></tr>
<tr><td></td><td>declaration__optional = declaration | "";</td><td>declaration__optional → declaration
declaration__optional → ε</td><td>declaration__optional(1: "INTEGER16") → declaration
declaration__optional(1: !"INTEGER16") → ε</td><td></td><td>declaration__optional = declaration | "";</td><td>{ LA_IS, { T_DATA_TYPE_0 }, { "declaration__optional",{\
{ LA_IS, {""}, 1, { "declaration" }}\
}}},\
{ LA_NOT, { T_DATA_TYPE_0 }, { "declaration__optional",{\
{ LA_IS, {""}, 0, { "" }}\
}}},\</td></tr>
<tr><td>value = [sign] , unsigned_value;</td><td>value = sign__optional, unsigned_value;</td><td>value → sign__optional unsigned_value</td><td>value(1: "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "+", "-") → sign__optional unsigned_value</td><td>value = -sign >> unsigned_value >> BOUNDARIES;</td><td>value = sign__optional >> unsigned_value >> BOUNDARIES;</td><td>{LA_IS, { "unsigned_value_terminal", T_ADD_0, T_SUB_0 }, { "value", {\
{LA_IS, { "" }, 2, { "sign__optional", "unsigned_value" }}\
}}},\</td></tr>
<tr><td></td><td>sign__optional = sign | ε;</td><td>sign__optional → sign
sign__optional → ε</td><td>sign__optional(1: "+", "-") → sign
sign__optional(1: !"+", !"-") → ε</td><td></td><td>sign__optional = sign | "";</td><td>{LA_IS, { T_ADD_0, T_SUB_0 }, { "sign__optional", {\
{LA_IS, { "" }, 1, { "sign" }}\
}}},\
{LA_NOT, { T_ADD_0, T_SUB_0 }, { "sign__optional",{\
{LA_IS, { "" }, 0, { "" }}\
}}},\</td></tr>
<tr><td>sign = sign_plus | sign_minus;</td><td>sign = sign_plus | sign_minus;</td><td>sign → sign_plus
sign → sign_minus</td><td>sign(1: "+") → sign_plus
sign(1: "-") → sign_minus</td><td>sign = sign_plus | sign_minus;</td><td>sign = sign_plus | sign_minus;</td><td>{LA_IS, { T_ADD_0 }, { "sign", {\
{LA_IS, {""}, 1, { "sign_plus" }}\
}}},\
{LA_IS, { T_SUB_0 }, { "sign", {\
{LA_IS, {""}, 1, { "sign_minus" } }\
}}},\</td></tr>
<tr><td>sign_plus = "+";</td><td>sign_plus = "+";</td><td>sign_plus → "+"</td><td>sign_plus(1: "+") → "+"</td><td>sign_plus = SAME_RULE(tokenPLUS);</td><td>sign_plus = SAME_RULE(tokenPLUS);</td><td>{LA_IS, { T_ADD_0 }, { "sign_plus", {\
{LA_IS, {""}, 1, {T_ADD_0}}\
}}},\</td></tr>
<tr><td>sign_minus = "-";</td><td>sign_minus = "-";</td><td>sign_minus → "-"</td><td>sign_minus(1: "-") → "-"</td><td>sign_minus = SAME_RULE(tokenMINUS);</td><td>sign_minus = SAME_RULE(tokenMINUS);</td><td>{LA_IS, { T_SUB_0 }, { "sign_minus", {\
{LA_IS, {""}, 1, {T_SUB_0} }\
}}},\</td></tr>
<tr><td>unsigned_value = non_zero_digit , {digit} | "0";</td><td>unsigned_value = non_zero_digit , digit__iteration | "0";</td><td>unsigned_value → non_zero_digit digit__iteration
unsigned_value → "0"</td><td>unsigned_value(1: "1", "2", "3", "4", "5", "6", "7", "8", "9") → non_zero_digit digit__iteration
unsigned_value(1: "0") → "0"</td><td>unsigned_value = (non_zero_digit >> *digit | digit_0) >> BOUNDARIES;</td><td>unsigned_value = (non_zero_digit >> digit__iteration | digit_0) >> BOUNDARIES;</td><td>/* <u>unsigned_value_token</u> represents <u>unsigned_value</u> in lexical analyzer */\
{LA_IS, { "unsigned_value_terminal" }, { "unsigned_value", {\
{LA_IS, { "" }, 1, { "unsigned_value_terminal" }}\
}}},\</td></tr>
<tr><td></td><td>digit__iteration = digit, digit__iteration | ε;</td><td>digit__iteration → digit digit__iteration
digit__iteration → ε</td><td>digit__iteration(1: "0", "1", "2", "3", "4", "5", "6", "7", "8", "9") → digit digit__iteration
digit__iteration(1: !"0", !"1", !"2", !"3", !"4", !"5", !"6", !"7", !"8", !"9") → ε</td><td></td><td>digit__iteration = digit >> digit__iteration | "";</td><td>\</td></tr>
</table>

<table>
<tr><td>digit = "0" | non_zero_digit;</td><td>digit = "0" | non_zero_digit;</td><td>digit → "0"
digit → non_zero_digit</td><td>digit(1: "0") → "0"
digit(1: "1", "2", "3", "4", "5", "6", "7", "8", "9",) → non_zero_digit</td><td>digit_0 = '0';
digit = digit_0 | non_zero_digit;</td><td>digit_0 = '0';
digit = digit_0 | non_zero_digit;</td><td>\</td></tr>
<tr><td>non_zero_digit = "1" | "2" | "3" | "4" | "5" | "6" | "7" | "8" | "9";</td><td>non_zero_digit = "1" | "2" | "3" | "4" | "5" | "6" | "7" | "8" | "9";</td><td>non_zero_digit → "1"
non_zero_digit → "2"
non_zero_digit → "3"
non_zero_digit → "4"
non_zero_digit → "5"
non_zero_digit → "6"
non_zero_digit → "7"
non_zero_digit → "8"
non_zero_digit → "9"</td><td>non_zero_digit(1: "1") → "1"
non_zero_digit(1: "2") → "2"
non_zero_digit(1: "3") → "3"
non_zero_digit(1: "4") → "4"
non_zero_digit(1: "5") → "5"
non_zero_digit(1: "6") → "6"
non_zero_digit(1: "7") → "7"
non_zero_digit(1: "8") → "8"
non_zero_digit(1: "9") → "9"</td><td>digit_1 = '1';
digit_2 = '2';
digit_3 = '3';
digit_4 = '4';
digit_5 = '5';
digit_6 = '6';
digit_7 = '7';
digit_8 = '8';
digit_9 = '9';
non_zero_digit = digit_1 | digit_2 | digit_3 | digit_4 | digit_5 | digit_6 | digit_7 | digit_8 | digit_9;</td><td>digit_1 = '1';
digit_2 = '2';
digit_3 = '3';
digit_4 = '4';
digit_5 = '5';
digit_6 = '6';
digit_7 = '7';
digit_8 = '8';
digit_9 = '9';
non_zero_digit = digit_1 | digit_2 | digit_3 | digit_4 | digit_5 | digit_6 | digit_7 | digit_8 | digit_9;</td><td>\</td></tr>
<tr><td>ident = "_" , letter_in_upper_case , letter_in_upper_case , letter_in_upper_case , letter_in_upper_case , letter_in_upper_case , letter_in_upper_case , letter_in_upper_case;</td><td>ident = "_" , letter_in_upper_case , letter_in_upper_case , letter_in_upper_case , letter_in_upper_case , letter_in_upper_case , letter_in_upper_case , letter_in_upper_case;</td><td>ident → "_" letter_in_upper_case
letter_in_upper_case letter_in_upper_case
letter_in_upper_case letter_in_upper_case
letter_in_upper_case letter_in_upper_case</td><td>Ident(1: "_") → "_" letter_in_upper_case
letter_in_upper_case letter_in_upper_case
letter_in_upper_case letter_in_upper_case
letter_in_upper_case letter_in_upper_case</td><td>tokenUNDERSCORE = "_";
ident = tokenUNDERSCORE >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> STRICT_BOUNDARIES;</td><td>tokenUNDERSCORE = "_";
ident = tokenUNDERSCORE >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> STRICT_BOUNDARIES;</td><td>/* <u>ident_token</u> represents <u>ident</u> in lexical analyzer */\
{LA_IS, { "ident_terminal" }, { "ident", {\
{LA_IS, {""}, 1, {"ident_terminal"} }\
}}},\</td></tr>
<tr><td>letter_in_lower_case = "a" | "b" | "c" | "d" | "e" | "f" | "g" | "h" | "i" | "j" | "k" | "l" | "m" | "n" | "o" | "p" | "q" | "r" | "s" | "t" | "u" | "v" | "w" | "x" | "y" | "z";</td><td>letter_in_lower_case = "a" | "b" | "c" | "d" | "e" | "f" | "g" | "h" | "i" | "j" | "k" | "l" | "m" | "n" | "o" | "p" | "q" | "r" | "s" | "t" | "u" | "v" | "w" | "x" | "y" | "z";</td><td>letter_in_lower_case → "a"
letter_in_lower_case → "b"
letter_in_lower_case → "c"
letter_in_lower_case → "d"
letter_in_lower_case → "e"
letter_in_lower_case → "f"
letter_in_lower_case → "g"
letter_in_lower_case → "h"
letter_in_lower_case → "i"
letter_in_lower_case → "j"
letter_in_lower_case → "k"
letter_in_lower_case → "l"
letter_in_lower_case → "m"
letter_in_lower_case → "n"
letter_in_lower_case → "o"
letter_in_lower_case → "p"
letter_in_lower_case → "q"
letter_in_lower_case → "r"
letter_in_lower_case → "s"
letter_in_lower_case → "t"
letter_in_lower_case → "u"
letter_in_lower_case → "v"
letter_in_lower_case → "w"
letter_in_lower_case → "x"
letter_in_lower_case → "y"
letter_in_lower_case → "z"</td><td>letter_in_lower_case(1: "a") → "a"
letter_in_lower_case(1: "b") → "b"
letter_in_lower_case(1: "c") → "c"
letter_in_lower_case(1: "d") → "d"
letter_in_lower_case(1: "e") → "e"
letter_in_lower_case(1: "f") → "f"
letter_in_lower_case(1: "g") → "g"
letter_in_lower_case(1: "h") → "h"
letter_in_lower_case(1: "i") → "i"
letter_in_lower_case(1: "j") → "j"
letter_in_lower_case(1: "k") → "k"
letter_in_lower_case(1: "l") → "l"
letter_in_lower_case(1: "m") → "m"
letter_in_lower_case(1: "n") → "n"
letter_in_lower_case(1: "o") → "o"
letter_in_lower_case(1: "p") → "p"
letter_in_lower_case(1: "q") → "q"
letter_in_lower_case(1: "r") → "r"
letter_in_lower_case(1: "s") → "s"
letter_in_lower_case(1: "t") → "t"
letter_in_lower_case(1: "u") → "u"
letter_in_lower_case(1: "v") → "v"
letter_in_lower_case(1: "w") → "w"
letter_in_lower_case(1: "x") → "x"
letter_in_lower_case(1: "y") → "y"
letter_in_lower_case(1: "z") → "z"</td><td>A = "A";
B = "B";
C = "C";
D = "D";
E = "E";
F = "F";
G = "G";
H = "H";
I = "I";
J = "J";
K = "K";
L = "L";
M = "M";
N = "N";
O = "O";
P = "P";
Q = "Q";
R = "R";
S = "S";
T = "T";
U = "U";
V = "V";
W = "W";
X = "X";
Y = "Y";
Z = "Z";
letter_in_lower_case = a | b | c | d | e | f | g | h | i | j | k | l | m | n | o | p | q | r | s | t | u | v | w | x | y | z;</td><td>A = "A";
B = "B";
C = "C";
D = "D";
E = "E";
F = "F";
G = "G";
H = "H";
I = "I";
J = "J";
K = "K";
L = "L";
M = "M";
N = "N";
O = "O";
P = "P";
Q = "Q";
R = "R";
S = "S";
T = "T";
U = "U";
V = "V";
W = "W";
X = "X";
Y = "Y";
Z = "Z";
letter_in_lower_case = a | b | c | d | e | f | g | h | i | j | k | l | m | n | o | p | q | r | s | t | u | v | w | x | y | z;</td><td>\</td></tr>
<tr><td>letter_in_upper_case = "A" | "B" | "C" | "D" | "E" | "F" | "G" | "H" | "I" | "J" | "K" | "L" | "M" | "N" | "O" | "P" | "Q" | "R" | "S" | "T" | "U" | "V" | "W" | "X" | "Y" | "Z";</td><td>letter_in_upper_case = "A" | "B" | "C" | "D" | "E" | "F" | "G" | "H" | "I" | "J" | "K" | "L" | "M" | "N" | "O" | "P" | "Q" | "R" | "S" | "T" | "U" | "V" | "W" | "X" | "Y" | "Z";</td><td>letter_in_upper_case → "A"
letter_in_upper_case → "B"
letter_in_upper_case → "C"
letter_in_upper_case → "D"
letter_in_upper_case → "E"
letter_in_upper_case → "F"
letter_in_upper_case → "G"
letter_in_upper_case → "H"
letter_in_upper_case → "I"
letter_in_upper_case → "J"
letter_in_upper_case → "K"
letter_in_upper_case → "L"
letter_in_upper_case → "M"
letter_in_upper_case → "N"
letter_in_upper_case → "O"
letter_in_upper_case → "P"
letter_in_upper_case → "Q"
letter_in_upper_case → "R"
letter_in_upper_case → "S"
letter_in_upper_case → "T"
letter_in_upper_case → "U"
letter_in_upper_case → "V"
letter_in_upper_case → "W"
letter_in_upper_case → "X"
letter_in_upper_case → "Y"
letter_in_upper_case → "Z"</td><td>letter_in_upper_case(1: "A") → "A"
letter_in_upper_case(1: "B") → "B"
letter_in_upper_case(1: "C") → "C"
letter_in_upper_case(1: "D") → "D"
letter_in_upper_case(1: "E") → "E"
letter_in_upper_case(1: "F") → "F"
letter_in_upper_case(1: "G") → "G"
letter_in_upper_case(1: "H") → "H"
letter_in_upper_case(1: "I") → "I"
letter_in_upper_case(1: "J") → "J"
letter_in_upper_case(1: "K") → "K"
letter_in_upper_case(1: "L") → "L"
letter_in_upper_case(1: "M") → "M"
letter_in_upper_case(1: "N") → "N"
letter_in_upper_case(1: "O") → "O"
letter_in_upper_case(1: "P") → "P"
letter_in_upper_case(1: "Q") → "Q"
letter_in_upper_case(1: "R") → "R"
letter_in_upper_case(1: "S") → "S"
letter_in_upper_case(1: "T") → "T"
letter_in_upper_case(1: "U") → "U"
letter_in_upper_case(1: "V") → "V"
letter_in_upper_case(1: "W") → "W"
letter_in_upper_case(1: "X") → "X"
letter_in_upper_case(1: "Y") → "Y"
letter_in_upper_case(1: "Z") → "Z"</td><td>a = "a";
b = "b";
c = "c";
d = "d";
e = "e";
f = "f";
g = "g";
h = "h";
i = "i";
j = "j";
k = "k";
l = "l";
m = "m";
n = "n";
o = "o";
p = "p";
q = "q";
r = "r";
s = "s";
t = "t";
u = "u";
v = "v";
w = "w";
x = "x";
y = "y";
z = "z";
letter_in_upper_case = A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z;</td><td>a = "a";
b = "b";
c = "c";
d = "d";
e = "e";
f = "f";
g = "g";
h = "h";
i = "i";
j = "j";
k = "k";
l = "l";
m = "m";
n = "n";
o = "o";
p = "p";
q = "q";
r = "r";
s = "s";
t = "t";
u = "u";
v = "v";
w = "w";
x = "x";
y = "y";
z = "z";
letter_in_upper_case = A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z;</td><td>\</td></tr>
<tr><td></td><td></td><td></td><td></td><td>STRICT_BOUNDARIES = (BOUNDARY >> *(BOUNDARY)) | (!(qi::alpha | qi::char_("_")));
BOUNDARIES = (BOUNDARY >> *(BOUNDARY) | NO_BOUNDARY);
BOUNDARY = BOUNDARY__SPACE | BOUNDARY__TAB | BOUNDARY__VERTICAL_TAB | BOUNDARY__FORM_FEED | BOUNDARY__CARRIAGE_RETURN | BOUNDARY__LINE_FEED | BOUNDARY__NULL;
BOUNDARY__SPACE = " ";
BOUNDARY__TAB = "\t";
BOUNDARY__VERTICAL_TAB = "\v";
BOUNDARY__FORM_FEED = "\f";
BOUNDARY__CARRIAGE_RETURN = "\r";
BOUNDARY__LINE_FEED = "\n";
BOUNDARY__NULL = "\0";
NO_BOUNDARY = "";
#define WHITESPACES \
STRICT_BOUNDARIES, \
BOUNDARIES, \
BOUNDARY, \
BOUNDARY__SPACE, \
BOUNDARY__TAB, \
BOUNDARY__VERTICAL_TAB, \
BOUNDARY__FORM_FEED, \
BOUNDARY__CARRIAGE_RETURN, \
BOUNDARY__LINE_FEED, \
BOUNDARY__NULL, \
NO_BOUNDARY</td><td>STRICT_BOUNDARIES = (BOUNDARY >> *(BOUNDARY)) | (!(qi::alpha | qi::char_("_")));
BOUNDARIES = (BOUNDARY >> *(BOUNDARY) | NO_BOUNDARY);
BOUNDARY = BOUNDARY__SPACE | BOUNDARY__TAB | BOUNDARY__VERTICAL_TAB | BOUNDARY__FORM_FEED | BOUNDARY__CARRIAGE_RETURN | BOUNDARY__LINE_FEED | BOUNDARY__NULL;
BOUNDARY__SPACE = " ";
BOUNDARY__TAB = "\t";
BOUNDARY__VERTICAL_TAB = "\v";
BOUNDARY__FORM_FEED = "\f";
BOUNDARY__CARRIAGE_RETURN = "\r";
BOUNDARY__LINE_FEED = "\n";
BOUNDARY__NULL = "\0";
NO_BOUNDARY = "";
#define WHITESPACES \
STRICT_BOUNDARIES, \
BOUNDARIES, \
BOUNDARY, \
BOUNDARY__SPACE, \
BOUNDARY__TAB, \
BOUNDARY__VERTICAL_TAB, \
BOUNDARY__FORM_FEED, \
BOUNDARY__CARRIAGE_RETURN, \
BOUNDARY__LINE_FEED, \
BOUNDARY__NULL, \
NO_BOUNDARY</td><td>\</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td><td>\
\</td></tr>
</table>

						\ { LA_IS, { T_NAME_0 }, { "program____part1",{\ { LA_IS, {""}, 7, { T_NAME_0, "program_name", T_SEMICOLON_0, T_BODY_0, T_DATA_0, "declaration_optional", T_SEMICOLON_0 }}\ }}},\ \ },\ "program_rule"